

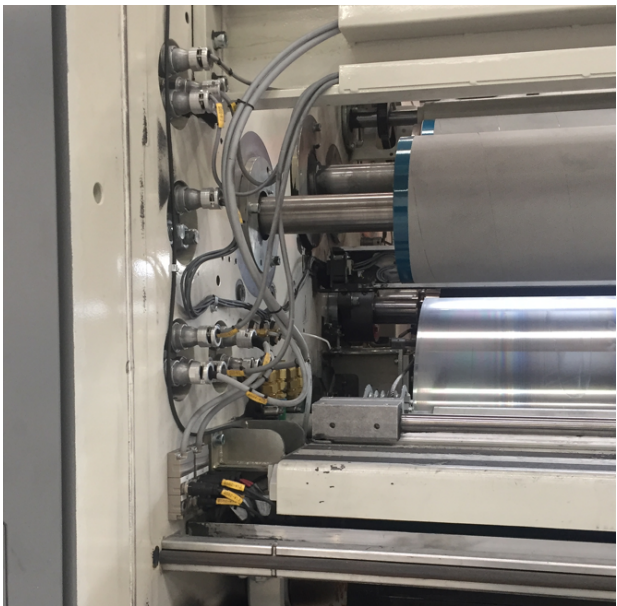
12.2.36 Chamber Viewing Windows

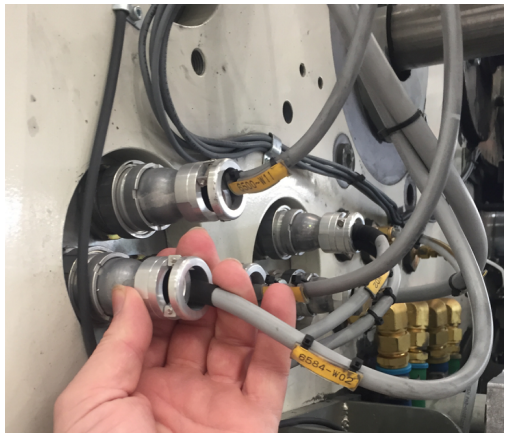
	Note !
	If Incorrect materials containing solvents or acids are used on or in the area of these chamber windows, there is a high possibility of failure

Maintenance Requirement	Equipment & Task to Be Performed
Daily	Visually check all windows for damage. Replace as required.
Figure 12.2-234 Example of a viewing window showing signs of “chemical attack”	

Maintenance Requirement	Equipment & Task to Be Performed
Daily	Visually check that the warning labels relating to Solvents are in place. If not replace ASAP
Figure 12.2-235 Example of a rear chamber window and its warning label	
Figure 12.2-236 Example of a front chamber window and its warning label	
Maintenance Requirement	Equipment & Task to Be Performed
Annually	Remove covers, if installed from the window fasteners and check for potential issues with cracks, etc. These will cause a vacuum leak and have to be overcome
Figure 12.2-237 Example of damaged window where excessive force has been applied to the locating bolts	

12.2.37 Electrical Lead Throughs – Vacuum

Maintenance Requirement	Equipment & Task to Be Performed
Weekly	Visually check for damage to cables, and repair/replace as necessary
Figure 12.2-238 Cables and leadthroughs at the winding cart faceplate	

Maintenance Requirement	Equipment & Task to Be Performed
Monthly	Check all lead-throughs for tightness, internally and externally
Figure 12.2-239 Example of a technician checking that the leadthrough plugs are fully engaged	

12.2.38 Process Gas Wedge –Automatic Flow Control System

Maintenance Requirement	Equipment & Task to Be Performed
Daily	Check gas wedge valve operation, with the machine under vacuum conditions. The web needs to be running to allow the technician to enable operate the flow from the process HMI screen.
Figure 12.2-240 Gas wedge flow control valves	

Maintenance Requirement	Equipment & Task to Be Performed
Monthly	Check that the MFC isolation valve solenoid is fully engaged and that the cable is not damaged.
Figure 12.2-241 Gas wedge flow control valves	

Maintenance Requirement	Equipment & Task to Be Performed
Monthly	Check that the cables and connectors are not damaged and are fully inserted in the device sockets
Figure 12.2-242 Gas wedge flow control valves	

Maintenance Requirement	Equipment & Task to Be Performed
Bi-Annual	Remove and clean the inlet filters using acetone. Ensure that they are fully dry before re-installing.
Figure 12.2-243 Position of the inlet filter	
Bi-Annual	Check the gas wedge is not blocked: - Remove the feed line to the gas wedge assembly at the chamber faceplate - Feed compressed air into the gas wedge assembly. If there are blockages, use a 1mm drill down the centre of the nylon nozzle Note: This task may need to be carried out more often if the machine is located in a dusty atmosphere.
Figure 12.2-244 Example of a technician removing the feed lines to the gas wedge assembly	

12.2.39 Siemens AC Vector Drive Motors

Maintenance Requirement	Equipment & Task to Be Performed
Monthly	Check and clean the cooling fan
Figure 12.2-245 Example of a technician checking the operation of the cooling fan	
Figure 12.2-246 Example of a technician checking the electrical connections to the motor drive "Cliq" communications system	
Figure 12.2-247 Example of a technician checking the electrical connections	 

Maintenance Requirement	Equipment & Task to Be Performed
Quarterly	Check encoder to motor connection is OK

Maintenance Requirement	Equipment & Task to Be Performed
Bi-Annually	Check the electrical connections for tightness inside the motors

Maintenance Requirement	Equipment & Task to Be Performed
Annually	Remove timing belts from belt driven assemblies and check for play in motor bearings
Annually	Carry out internal bearing analysis, if relevant equipment is available.

12.2.40 Winding System Drive Belt Access Covers/Guards

Maintenance Requirement	Equipment & Task to Be Performed
Weekly	Check that all guards are in position and that locating fasteners are in place

Maintenance Requirement	Equipment & Task to Be Performed
Figure 12.2-248 The winding cart side panel	
	 Locating bracket
	 Base Locking Screw



Warning !

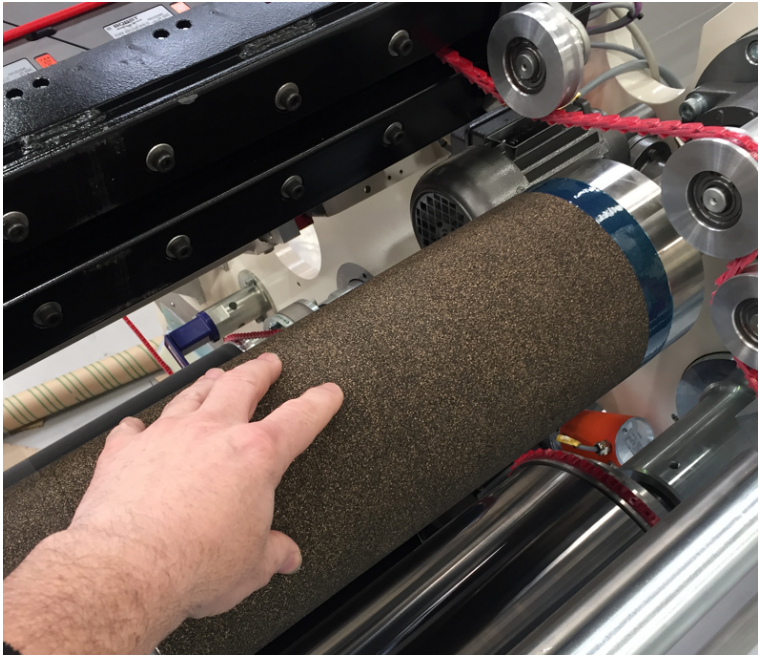
Do not operate this equipment without the guards being in place and access doors closed.

Maintenance Requirement	Equipment & Task to Be Performed
Weekly	Check that the access door to the winding cart lock is operational.

Maintenance Requirement	Equipment & Task to Be Performed
Figure 12.2-249 The winding cart access door	

Maintenance Requirement	Equipment & Task to Be Performed
Figure 12.2-250 The winding cart fixed panel external and internal fixing points	
Weekly	Check that the appropriate safety warning signs have not been removed. If unsure contact Bobst Manchester for advice.

12.2.41 Bowed Spreader Roller - Kickert

Maintenance Requirement	Equipment & Task to Be Performed
Weekly	Check the rubber sleeve for damage
Weekly	Check that the apex position on the roller is represented correctly on the operator HMI drives page
Weekly	Check the operation of the apex position motor in both clockwise and counter clockwise directions.
Weekly	Rotate the roller by hand and check for uneven rotation. If uneven, this could be a indication that the roller requires overhauling.
Figure 12.2-251 Example of a technician checking the condition of rubber sleeve	

Maintenance Requirement	Equipment & Task to Be Performed
Monthly	Check the location fasteners on the roller

Maintenance Requirement	Equipment & Task to Be Performed
5 Years	Send roller back to manufacturer for roller overhaul.

12.2.42 Winding Cart Rotary Vacuum Lead Throughs – High Speed

Maintenance Requirement	Equipment & Task to Be Performed
Daily	Visually check the gravity oil bottles for loss of oil. Major losses will indicate seal failure.
Daily	Visually check the gravity oil lines for indication of oil. Repair as necessary

Maintenance Requirement	Equipment & Task to Be Performed
Weekly	Check internal shaft for indication of major oil leak. It can be expected that a small amount of oil will be present, that lubricates the seal in operation

Figure 12.2-252 Example of a technician checking the gravity oil bottles



Maintenance Requirement	Equipment & Task to Be Performed
Bi-Annual	Check for bearing play In all rotary lead-throughs

Maintenance Requirement	Equipment & Task to Be Performed
Annual	Check for bottle dis-colourment. Replace if dis-coloured.

12.2.43 Advanced Energy Plasma Power Supply

On some contracts, the plasma treater power supply is located in a separate electrical panel.



Front
Access
View

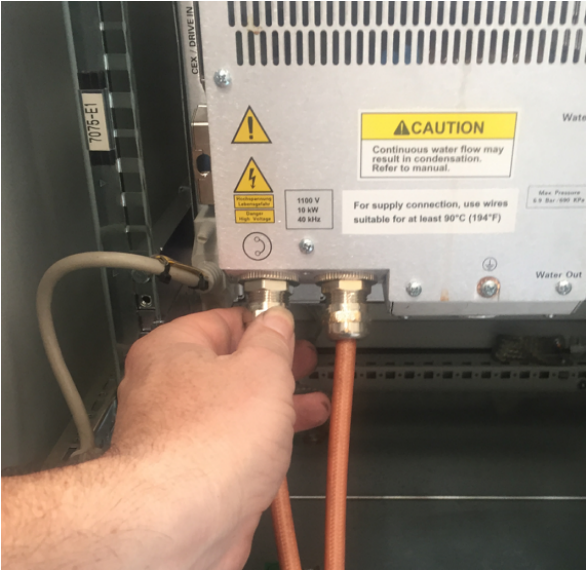


Rear
Access
Panel

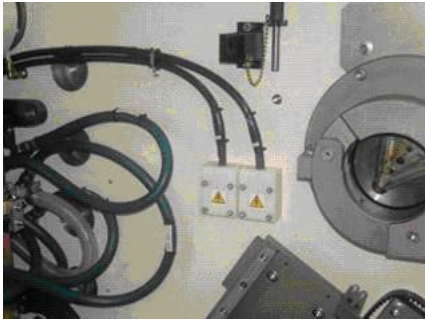
Figure 12.2-253 E07_0 electrical panel containing the power supply

Maintenance Requirement	Equipment & Task to Be Performed
Monthly	Check that water cooling lines to the power supply are not damaged or show indication of condensation
Figure 12.2-254 Example of a technician checking the internal water cooling	

Maintenance Requirement	Equipment & Task to Be Performed
Monthly	Clean/replace filter material in the rear panel door.
Figure 12.2-255 Example of a technician removing external filter cover	

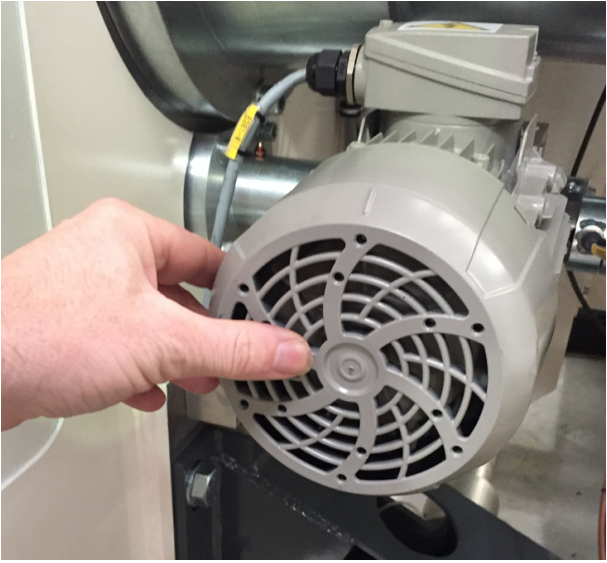
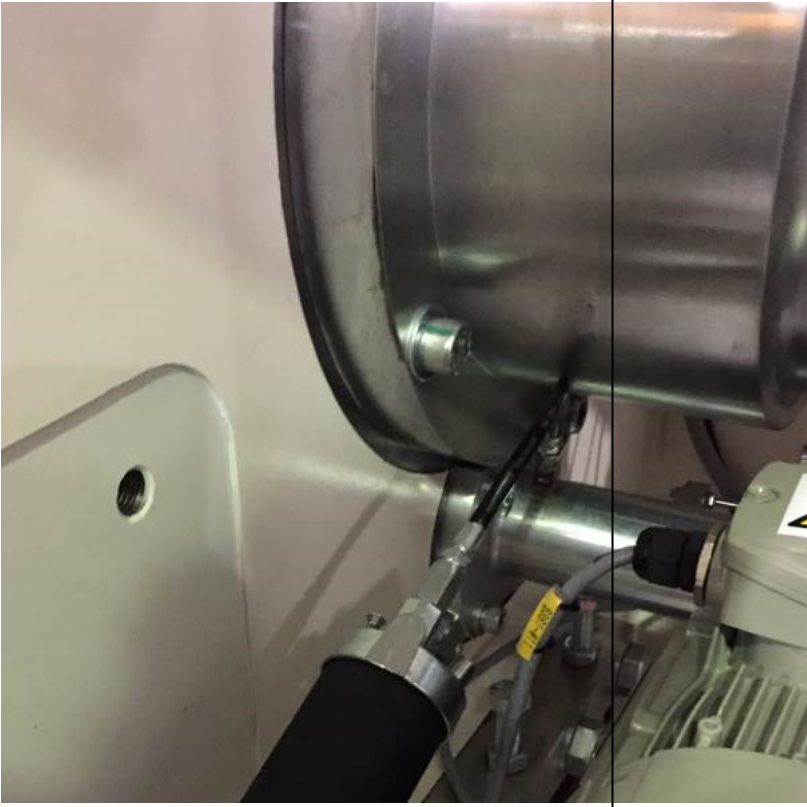
Maintenance Requirement	Equipment & Task to Be Performed
Monthly	Check that internal cables are secure and show no indication of physical damage or overheating
Figure 12.2-256 Example of a technician checking the condition of the high voltage cable	
Figure 12.2-257 Example of a technician checking the condition of the power supply cable	

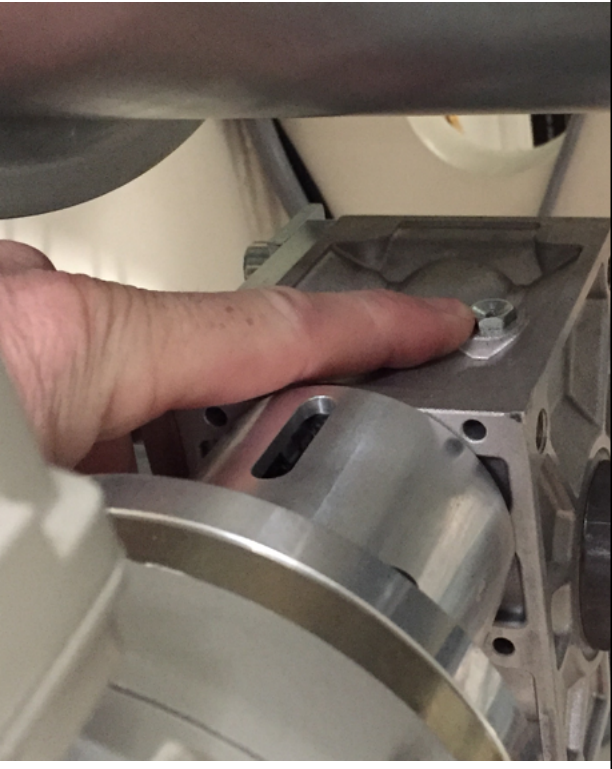
12.2.44 Plasma High Voltage Interconnection Cables

Maintenance Requirement	Equipment & Task to Be Performed
Quarterly	Visually check for damage to the high voltage cables.
Quarterly	Check the connections for tightness and damage, internally and externally
Figure 12.2-258 Plasma supply internal connections	
Figure 12.2-259 Plasma supply external connections	

12.2.45 Chuck Sidelay - Screw Jack

Maintenance Requirement	Equipment & Task to Be Performed
Monthly	Check operation of jack through full movement, using the sidelay buttons at the rear HMI station Check the limit switch wiring is fully engaged and the system works within these limits.
Figure 12.2-260 Screw jack assembly	

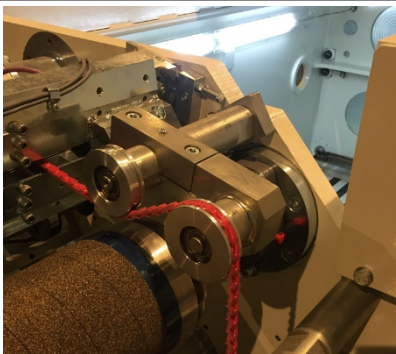
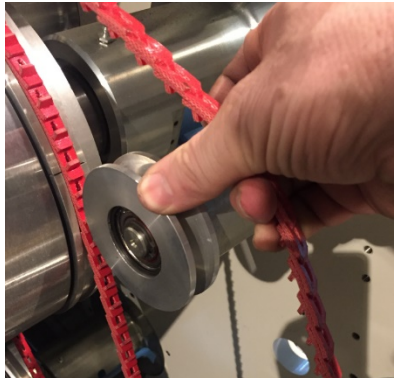
Maintenance Requirement	Equipment & Task to Be Performed
Quarterly	Wipe motor fan body to ensure correct cooling taking place.
Figure 12.2-261 Example of a technician checking and cleaning motor body	
Quarterly	Grease the rod leadthrough
Figure 12.2-262 Example of a technician greasing the leadthrough	

Maintenance Requirement	Equipment & Task to Be Performed
Quarterly	Check the internal lip seal, and condition of rod. There should be no damage to either.
Figure 12.2-263 Example of a technician checking the condition of the internal lip seal	 A close-up photograph of a mechanical assembly. A metal rod is inserted into a circular housing. A person's finger is pointing to the internal lip seal where the rod meets the housing. The seal appears to be a black O-ring or similar component.
Quarterly	Grease the gearbox in accordance with the manufacturers recommendations
Figure 12.2-264 Example of a technician identifying the grease nipple point on the gearbox	 A close-up photograph of a metal gearbox. A person's finger is pointing to a small, hexagonal grease nipple on the side of the gearbox. The gearbox is made of cast metal and has various ports and openings.

Maintenance Requirement	Equipment & Task to Be Performed
Quarterly	Visually check the condition of the ball screw, through the hole in the cover.
Figure 12.2-265 Example of a technician checking the condition of the ball screw	

12.2.46 Winding Mechanism – Web threader

Maintenance Requirement	Equipment & Task to Be Performed
Weekly	Check the condition of the belt, and replace as necessary
Figure 12.2-266 Example of a technician checking the condition of the threader belt	
Weekly	Check the tension applied to the belt, to ensure that it is not excessive.
Weekly	Check the belt for any build-up of tape. This should be removed

Maintenance Requirement	Equipment & Task to Be Performed
Monthly	Check that all pulleys are free to rotate
Figure 12.2-267 Belt guidance pulleys	
Figure 12.2-268 Example of a technician checking the pulley rotation	

Maintenance Requirement	Equipment & Task to Be Performed
Annually	Replace the T belt

12.2.47 Gearboxes

There are some gearboxes on this machine, which have been supplied lubricated for the life of the component.

Maintenance Requirement	Equipment & Task to Be Performed
Weekly	Check all gearboxes for excessive noise (where possible)

Maintenance Requirement	Equipment & Task to Be Performed
Monthly	Check all gearboxes for indication of leakage and resolve any found.

Maintenance Requirement	Equipment & Task to Be Performed
Annual	All gearboxes should have their oil or grease changed, where possible. See the manufacturers manual for details.

12.3 Lubrication

The following pages contain reference tables of the oils and greases used in the various components used on Bobst Manchester Ltd. machines.

	Note !
	Contaminated components and operating media must be disposed of in accordance with local government regulations where the machine is operated.

	Warning !
	Pump oil and oil mists contaminated with hazardous substances can be hazardous to your health. Skin or eye contact with pump oil and inhaling oil mists can be hazardous to your health. Wear relevant personal protective equipment when draining oil from vacuum pumps, which can potentially contain harmful gases or other harmful substances, and when disposing of exhaust condensation or filters.

	Caution !
	The lubricating oils from the vacuum pumps are potentially harmful to the environment. Never allow pump oil or parts dipped in pump oil to penetrate the supply of drinking water. Pump oil (even small amounts!) must be disposed of as hazardous waste.

Lubrication Tasks

The following table shows the suggested grease lubrication period on various equipment found on the machine. Photographs showing the actual greasing points are shown in the preventative maintenance schedule or in the vendor manuals supplied.

	Caution !
	Failure to follow recommended preventative maintenance procedures may lead to machinery damage. Do not use oil or grease types that are not on the recommended list, as this could lead to failure of components upon the machine.

Grease Lubrication

Equipment	Grease Type	Daily	Weekly	Monthly	Quarterly	Bi-Annual	Annual	Removal	Strip down
Deublin Rotary Gland/Union	I		✓					✓	

Table 12.3-1 Lubrication Schedule - Greases

Equipment	Grease Type	Daily	Weekly	Monthly	Quarterly	Bi-Annual	Annual	Removal	Strip down
Face-plate O-ring	C							✓	
Winding Mech. ABB Load cell bearings	A							✓	
Winding Cart Movement Castors	A			✓				✓	
Coating Shield Pivot Internal Bearing	A			✓				✓	
Winding Mechanism Roller Bearings	A		✓					✓	✓
Shutter Lubrication - Cams	A							✓	
Reel Shaft Bearings	A				✓			✓	
Shutter Drive Bearings	A							✓	
Vacuum Test Ports	C							✓	✓
Vacuum Pipework O-rings	C							✓	✓
Chamber Window O-rings	C							✓	✓
Vacuum Pipework Isolation Valve Seals	C							✓	✓
Vacuum Pipework Rubber Bellows	A							✓	✓
Grease Filled Gearboxes- inside chamber	A							✓	✓
Layarm Slides (grease type)	A		✓						
Grease Filled Gearboxes external	see plates								
Strobe Cylinder Bearings	A			✓				✓	✓
Kickert Spreader Roller	A			✓				✓	✓
Kickert Spreader Roller Gearbox	A			✓				✓	✓

Table 12.3-1 Lubrication Schedule - Greases

Equipment	Grease Type	Daily	Weekly	Monthly	Quarterly	Bi-Annual	Annual	Removal	Strip down
Evaporation Source Flap Rotation	A			✓				✓	✓
Internal Castor Bearings	A			✓				✓	
Transformer to Steel Connections	F								✓
Unwind layarm leadthrough	A			✓				✓	✓
Electrical Leadthrough – Gaskets	C							✓	✓
Unwind Chuck Sidelay – Linear Slide (option)	A			✓				✓	✓

Table 12.3-1 Lubrication Schedule - Greases

Key to above chart:

D Daily
 M Monthly
 BA Bi-annual
 R Removal Of Equipment For Repair S Strip Out and replacement
 W Weekly
 Q Quarterly
 A Annual

	Note !
	 - The majority of assembly drawings supplied will refer to the machine lubrication drawing, which can be located on the electronic CD and this section of the manual. - Grease types are described in other sections of this manual. - Please note that cross-reference of the greases and the codes used by suppliers may change from country to country. - If users propose to use a grease type not listed, it is suggested that before the grease is purchased, the specification of the grease is to be sent to the equipment manufacture for their recommendations. - The greases used within the machine are selected for the required working conditions and low outgassing properties.

Oil Lubrication

Equipment	Daily	Weekly	Monthly	Quarterly	Bi-Annual	Annual	Removal	Strip down
Busch dry pumps						✓		✓
Aerzen mechanical booster pumps					✓	✓	✓	

Table 12.3-2 Lubrication Schedule – Oil

Equipment	Daily	Weekly	Monthly	Quarterly	Bi-Annual	Annual	Removal	Strip down
Diffusion pumps						✓		✓
Unwind chuck gravity bottle							✓	✓
Draw							✓	✓
Rewind chuck gravity bottle								
Drum leadthrough gravity bottle							✓	✓
Draw roller leadthrough gravity bottle							✓	✓
DTR1 roller leadthrough gravity bottle							✓	✓
DTR2 roller leadthrough gravity bottle							✓	✓
Viewing flap valve leadthrough gravity bottle							✓	✓
Shutter drive leadthrough gravity bottle								
High vacuum valve (VV7) leadthrough gravity bottle								
Chiller Compressor							✓	✓
Brookes Maxcool Unit Compressor							✓	✓
Lauda Refrigeration System								✓
Winding Cart Movement Gearbox						✓	✓	✓

Table 12.3-2 Lubrication Schedule – Oil

Replacement Parts

Any replacement parts such as load cell transducer bearings must be cleaned with a degreaser and then acetone before being re-greased with the approved grease type.

Lubrication Points on Machine

Refer to the preventative maintenance section of the manual for photographs of lubrication tasks taking place.

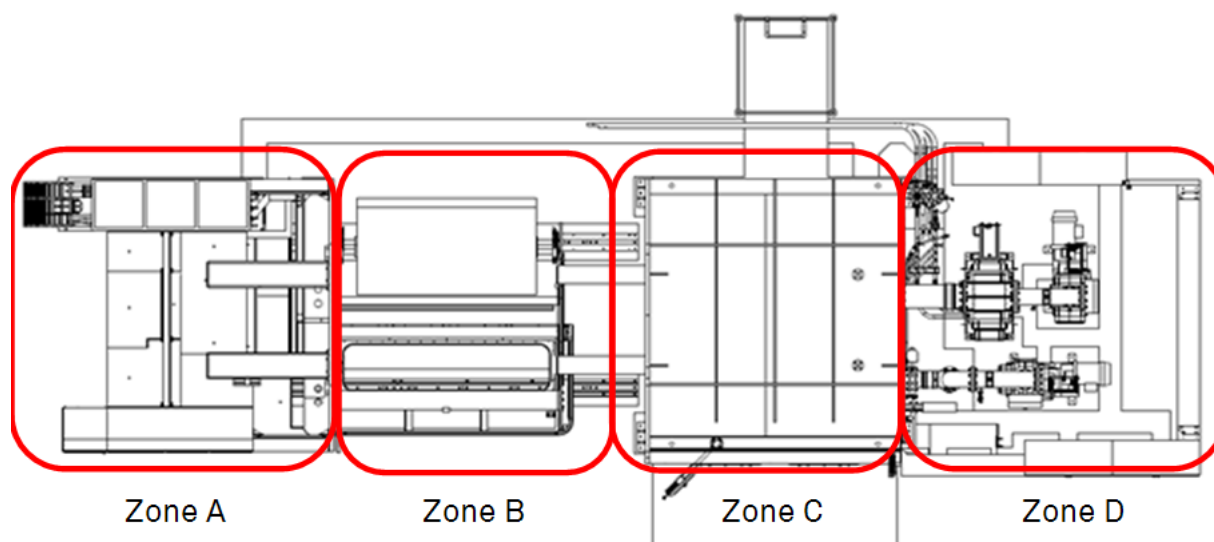


Figure 12.3-1 Location of Equipment Requiring Oil Lubrication

Equipment	Grease Type	Machine Location zone
Deublin Rotary Union	I	A
Face-plate O-ring	C	B
Winding Mech. ABB Load cell bearings	A	B
Winding Cart Movement Castors	A	A
Coating Shield Pivot Internal Bearing	A	B
Winding Mechanism Roller Bearings	A	B
Shutter Lubrication - Cams	A	B
Reel Shaft Bearings	A	B
Shutter Drive Bearings	A	B
Vacuum Test Ports	C	A/C/D
Vacuum Pipework O-rings	C	D
Chamber Window O-rings	C	C
Vacuum Pipework Isolation Valve Seals	C	D
Vacuum Pipework Rubber Bellows	A	D
Grease Filled Gearboxes- inside chamber	A	B
Grease Filled Gearboxes external	see plates	A
Viewing Strobe Cylinder Bearings	A	C Internal
Kickert Spreader Roller	A	B
Internal Castor Bearings	A	B

Table 12.3-3 Lubrication Schedule – Greases

Equipment	Grease Type	Machine Location zone
VV7 Valve Leadthroughs – Grease Type	A	C External
Unwind layarm leadthrough	A	A
Electrical Leadthrough – Gaskets	C	A/C/D

Table 12.3-3 Lubrication Schedule – Greases

Equipment	Oil Type	Machine Location zone
Bush Dry Pumps	See chapter Lubrication Oils below	D
Aerzen Mechanical Booster Pumps	See chapter Lubrication Oils below	D
Diffusion pumps	See chapter Lubrication Oils below	C (Under chamber)
Unwind Chuck Gravity Bottle	See chapter Lubrication Oils below	A
Draw Roller Gravity Bottle	See chapter Lubrication Oils below	A
Rewind Chuck Gravity Bottle	See chapter Lubrication Oils below	A
Drum Leadthrough Gravity Bottle	See chapter Lubrication Oils below	A
DTR1 Leadthrough Gravity Bottle	See chapter Lubrication Oils below	A
DTR2 Leadthrough Gravity Bottle	See chapter Lubrication Oils below	A
Flap Valve Leadthrough Gravity Bottle	See chapter Lubrication Oils below	D
Chiller Compressor	See chapter Lubrication Oils below	A
Brookes Polycold Compressor	See chapter Lubrication Oils below	D
Winding Cart Movement Gearbox	See chapter Lubrication Oils below	A

Table 12.3-4 Lubrication Schedule – Oil

Equipment Oil /Grease Types

	Caution !
	Do not use oil types that are not on the recommended list, as this could lead to failure of components upon the machine.

Lubrication Oils

Equipment	Busch Screw Vacuum Pump – Cobra NC 630 B
Capacity:	See Manufactures Documentation
Special Notes:	
Recommended Lubricant Type from vendor	Anderol 555

Equipment	Aerzen Mechanical Booster Pumps - Canned
Capacity:	Size GM 2000 2.5 litres / GL 8000 7.6 litres
Special Notes:	
Recommended Lubricant Type from vendor	Anderol 555

Equipment	Varian Diffusion Pumps
Capacity:	NHS 20 – 4.7 litres
Special Notes:	
Recommended Lubricant Type from vendor	Bobst CVC Silicone 4

Equipment	Non-cooled gravity Oil Lubrication Gravity Feed Bottles
Capacity:	Dependent upon chuck size
Special Notes:	Fill within 11 mm of the top of the gravity bottle.
Recommended Lubricant Type from vendor	BML HYD10
Equivalent Type 1	Mobil (UK)DTE Heavy Medium

	Note !
	Unwind/Rewind/Draw and DTR2 leadthroughs

Equipment	Cooled Drum/rollers gravity Oil Lubrication Gravity Feed Bottle
Capacity:	Dependent upon leadthrough size
Special Notes:	Fill within 11 mm of the top of the gravity bottle. Only to be used on leadthroughs that are cooled i.e. drum leadthrough
Recommended Lubricant Type from vendor	Castrol(UK) Hyspin AWS 10

	Note !
	Process Drum (WCD) and DTR1(WCR) leadthroughs

Equipment	Process Chiller Compressor Oil
Capacity:	See Vendor manual
Special Notes:	For R404 Gas
Recommended Lubricant Type from vendor	Dependent upon refrigerant gas charge type

Equipment	Cryo-generator Compressor Oil
Capacity:	3.9 litres
Special Notes:	see vendor manual
Recommended Lubricant Type from vendor	Solest LT-32

Equipment	Traverse Movement Gearbox Oil
Capacity:	Varies , see vendor manual
Special Notes:	
Recommended Lubricant Type from vendor	BP(UK) Energol GR-XP 220
Equivalent Type 1	Esso Spartan EP 220
Equivalent Type 2	Shell(UK) Omala 220
Equivalent Type 3	Mobil(UK) Gear 630

Table 12.3-5 Lubrication Oil Details

	Caution !
	Do not use lubricant types that are not on the recommended list, as this could lead to failure of components upon the machine.

Gearbox Capacities and Oil Level Indication Capacities

See documentation for individual capacities in the equipment data/manuals on the machine documentation CD/DVD.

Oil Levels

The equipment supplied has some vendor guidelines in regards to oil levels. Please refer to the vendor manuals supplied.

In the case of the gravity bottles, the oil level should be above the half way part of it

Lubrication Greases

Grease Type:	A
Recommended Lubricant Type from vendor	Mobil (US) Grease 28
Equivalent Type 1	Mobil (US) Supertemp SHC32
Equivalent Type 2	Castrol (UK) Longtime PD2

Grease Type:	A
Equivalent Type 3	Esso(UK) Nova 275
Equivalent Type 4	Shell(US) Retina AX

Grease Type:	B
Recommended Lubricant Type from vendor	Dow Corning Molykote 44M
Equivalent Type 1	Shell(UK) Alvania RA
Equivalent Type 2	Mobil(US) Mobilplex 47
Equivalent Type 3	Esso(UK) Beacon 2
Equivalent Type 4	BP(UK) Energrease LS2

Grease Type:	C
Recommended Lubricant Type from vendor	Apiezon M Grease
Equivalent Type 1	Dow Corning High Vacuum Grease
Equivalent Type 2	Dupont Krytox LVP2
Equivalent Type 3	Varian Vacuum Grease

Grease Type:	D
Recommended Lubricant Type from vendor	Shell(UK) Aero Shell #7
Equivalent Type 1	Esso(UK) Temperex N3
Equivalent Type 2	SKF LGHQ3

Grease Type:	E
Recommended Lubricant Type from vendor	Shell Dolium R
Equivalent Type 1	Chevron SRI

Grease Type:	F
Recommended Lubricant Type from vendor	Tecknit Conductive Silver Filled silicone Grease

Grease Type:	G
Recommended Lubricant Type from vendor	Kluber HQ 72-72

Grease Type:	H
Recommended Lubricant Type from vendor	Kluber F-V4

Grease Type:	I
Special Notes:	
Recommended Lubricant Type from vendor	Kluber Petamo GHY 113N

Table 12.3-6 Lubrication Grease Details

	Note !
	- There may be other gearboxes fitted to components on the equipment. They are normally labelled with their requirements. If no details are available, please contact Bobst Manchester Ltd - The code/names given by lubricant manufacturers may change between continents.

Grease and Oil Specifications & MSDS Information

The following list references the specifications on some of the common lubricants that are used in the machines.

The table is used to identify the actual lubricants.

The term MSDS is a short form for material safety data sheet.

Where possible BOBST has collected these together and have supplied a copy of these documents within the electronic media provided.

Manufacturer	Product	Data Sheet Available from BML	MSDS Available from BML	Applicable to this machine
Anderol	555 type oil	✓	✓	✓
Apeizon	M Vacuum Grease	✓	✓	✓
Apollo	Type 32 Hydraulic oil	N/A	✓	✓
BP	Energol GR-XP 220	✓	✓	
BOBST	CVC4 – Diffusion Pump Oil	✓	✓	✓
Castrol	Hyspin 10	✓	✓	✓
	Tribol 800 Oil	✓		
	Longtime PD2 Grease	✓	✓	✓
Chevron	Type SRI Grease	✓	✓	
Crown	Hydraulic Oil	N/A	✓	
	Molykote 44 Grease	✓	✓	✓
	Type 55 Grease	✓	✓	✓
	Type 200 Oil	✓	✓	

Manufacturer	Product	Data Sheet Available from BML	MSDS Available from BML	Applicable to this machine
	High Vacuum Grease		✓	✓
Esso	Spartan EP Oil	✓	✓	
Kluber	Petamo GHY 113N	✓	✓	
	BQ 72-72 Grease	✓		
	Microlube GB0	N/A	✓	
	Oil Data sheet	✓		
Mobil	Mobilgear 630 Oil	✓		
	Mobilgrease 28 grease	✓	✓	✓
	DTE Oil	✓	N/A	
	DTE Oil Heavy Medium MSDS	N/A	✓	✓
	EAL Artic 22	✓	✓	✓
	Nuto H Hydraulic Oil	✓		
	SHC Series Data	✓	✓	✓
	SHC 32 Grease	✓	✓	✓
Polycold	Soltest 32 Oil	N/A	✓	✓
Shell	Clavus Oil	✓	✓	
	Corena Oil	✓	✓	
	Dolium BRB Grease	✓	✓	
	Retina Grease	✓	✓	
	AW Hydraulic 32 Oil	✓	✓	✓
	Aeroshell 7	✓	✓	
	Omala HD320 Oil	✓	✓	
	Tellus 68 Oil	✓	✓	
	Turbo Oil	✓	✓	
SKF	LGHQ3 Grease	✓	✓	
Tecknit	Conductive Silicone Grease	✓	✓	✓
Total	Diekan 1640 Oil	✓	✓	✓